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Sri Lanka Institute of Information Technology

BSc (Hons) in Information Technology Specializing in Cyber Security
IT4010 - Research Project (Comprehensive Design and Analysis Project)

R26-CS-010

Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training

Progress Logbook

IT number	Student Name
IT22309556	PRIYADARSHI S. A. S. D.

Date: 20th Feb - 11th Mar

Individual Component: Privacy Integration: Researching mathematical privacy models and enforcing privacy during data generation.

Summary of RP work carried out during the reporting period

Date	RP Work
20/02/2026	Read initial literature on mathematical privacy models applicable to machine learning and autonomous vehicle datasets.
28/02/2026	Focused research on ϵ-differential privacy (ϵ-DP) principles and its mathematical formulation for generative models.
04/03/2026	Investigated methodologies for enforcing ϵ-DP *during* the synthetic data generation phase, rather than relying on post-processing anonymization.
08/03/2026	Reviewed and compared specific privacy-preserving machine learning libraries, focusing on Opacus (for PyTorch) Privacy.
11/03/2026	Finalized library selection based on the team's baseline utility model and documented the theoretical integration plan for the data pipeline.

Supervisor's comments.

Student progressing well in research project: Yes / No

If NO:

Project work to be carried out before next evaluation period:

Signature: 

Date: 13/03/2026

Date: 15th Mar - 25th Apr

Individual Component: Privacy Integration: Implementing the ε-differential privacy enforcement pipeline including the Privacy-Aware Calibration Module, DP Training Wrapper, Privacy Budget Accountant, and novel epsilon calibration attack study.

Summary of RP work carried out during the reporting period

Date	RP Work
15/03/2026	Began Phase 2 implementation. Set up Conda virtual environment (dp_av_env) with Python 3.11.15 using Miniconda on Windows 11. Resolved Python 3.14 incompatibility with PyTorch wheel distributions.
17/03/2026	Installed PyTorch 2.5.1 with CUDA 12.1 build, verified GPU availability (NVIDIA GeForce RTX 3050 Ti Laptop GPU, 4GB VRAM). Ran initial smoke test confirming Opacus 1.4.1 installs cleanly alongside PyTorch.
19/03/2026	Confirmed Opacus RDP accountant functionality via smoke test: trained toy model for 3 epochs, verified cumulative ε tracked correctly per step, budget log exported as CSV (48 rows, one per training step). All FR-01 through FR-07 behaviors confirmed at framework level.
21/03/2026	Began implementing the Privacy-Aware Calibration Module. Designed the LaplaceCalibrationModule class structure: privatise_scalar() for numerical aggregates and privatise_histogram() for distributional statistics, with per-query audit logging.
23/03/2026	Completed LaplaceCalibrationModule implementation. Laplace noise applied as: noise = Laplace(0, sensitivity / ε). Histogram bins clipped to zero and renormalized after noise addition to preserve valid probability distributions. Calibration ε hardcoded below 2.0 (20% of total budget) with ValueError on violation.

25/03/2026	Ran Privacy-Aware Calibration Module against simulated Waymo aggregate statistics. Generated calibration_stats.json containing seven privatized fields: vehicle density, pedestrian density, vehicle speed, pedestrian speed, intersection density, weather distribution, and time-of-day distribution. Confirmed ε_calibration = 1.8 and calibration_audit_log.json written correctly.
27/03/2026	Sent calibration_stats.json and interface specification document to IT22541284. Defined field schema, Laplace noise parameters, sensitivity values, and CARLA Python API mapping for each field. Clarified fog bin zero-value behavior (Laplace noise on small bins rounds to 0.0 after clipping — expected, not a bug).
29/03/2026	Began implementing the DP Training Wrapper. Integrated Opacus PrivacyEngine with PyTorch using make_private_with_epsilon() for automatic noise multiplier computation. Implemented BudgetExceededError halt mechanism: training loop terminates automatically when RDP accountant reaches configured ε ceiling.
01/04/2026	Completed DPTrainingWrapper class. Per-sample gradient clipping enforced at L2 norm C=1.0. Gaussian noise injected at scale C × σ before optimizer weight update. Opacus ModuleValidator.fix() auto-replaces BatchNorm layers with GroupNorm for compatibility. FR-02 and FR-03 confirmed via unit-level inspection of gradient norms.
03/04/2026	Ran three ablation configurations as specified in proposal: Config A (ε=1.0, σ=1.97), Config B (ε=3.0, σ=1.01), Config C (ε=8.0, σ=0.67). All three runs completed 125 steps over 5 epochs. Confirmed noise multiplier scales inversely with target ε as expected from RDP composition theory.

05/04/2026	Verified FR-06 and FR-07 compliance across all three ablation configs. Per-step budget logs saved as CSV (125 rows each, columns: step, cumulative_eps, loss). Structured audit reports generated in JSON format containing final ε, clipping norm C, noise multiplier σ, total steps, halted_early flag, and budget_respected flag. Reports forwarded to IT22066916.
07/04/2026	Identified novel contribution opportunity: empirical calibration between formal ε bounds and actual membership inference attack (MIA) resistance. No prior published work has produced this curve for multi-modal AV synthetic data. Framed research question: does formal ε predict real-world attack resistance in the AV domain?
09/04/2026	Designed and implemented the ShadowMIAAttack class following the four-stage shadow model protocol from Shokri et al. (2017): • train N shadow models mimicking target model behavior. • collect confidence score profiles from member and non-member samples. • train binary attack classifier on profiles. • apply classifier to actual target model and measure success rate.
11/04/2026	Extended attack beyond Shokri et al. (2017) with a two-feature signal design: confidence score and per-sample loss computed jointly for each sample. Rationale: for generative models, per-sample loss captures gradient-level memorization more directly than output confidence alone. Attack classifier input dimension updated from 1 to 2. Validated that member samples show substantially lower loss than non-members in shadow model training (mean member loss ≈0.36 vs non-member ≈0.88).
13/04/2026	Ran full epsilon calibration study: attacked models trained under Config A (ε=1.0), Config B (ε=3.0), Config C (ε=8.0), and a no-DP baseline. All configurations produced attack success rates near 50% on toy tensor data, confirming that random tensor datasets do not exhibit memorization. Framework validated end-to-end. Real calibration curve will be produced in Phase 3 against CARLA generative model outputs.

15/04/2026	Calibration curve data saved to calibration_curve_data.json with description, random baseline (50%), target ceiling (55%), and data points per config. Documented novel contribution framing for thesis: first empirical ε-to-attack-resistance calibration study for multi-modal AV synthetic data.
17/04/2026	Coordinated with IT22066916: shared shadow_mia_attack.py and run_calibration_study.py, briefed on Phase 3 attack protocol. Confirmed that audit reports from the DP Training Wrapper (final ε, C, σ) will be passed to IT22066916 before MIA testing begins in Phase 3.
19/04/2026	Responded to IT22110220 query regarding privacy-modified image output. Clarified that this project enforces privacy at the gradient level during training (DP-SGD via Opacus), not by applying noise or blurring to output images. Synthetic images produced by CARLA will appear visually normal; the privacy guarantee is quantified by the RDP accountant log and audit reports.
21/04/2026	Generated placeholder synthetic dataset for IT22110220's utility evaluation pipeline setup: camera tensor (200, 3, 256, 256), LiDAR tensor (200, 1024, 4), and ground-truth label tensor (200, 10, 5) across all three epsilon configs. Metadata JSON included with field descriptions, epsilon value, and audit report reference. Enables IT22110220 to build and test her training pipeline before real CARLA data is available.
23/04/2026	Reviewed Waymo Open Dataset access requirements. Confirmed existing access covers the aggregate statistical extraction needed for Phase 3 calibration. Designed pseudocode for extract_waymo_stats.py: in-memory processing of raw sensor segments to extract vehicle counts, pedestrian counts, weather distributions, and time-of-day distributions without writing raw frames to disk.

25/04/2026	Completed Phase 2 documentation and review. All four sub-modules operational: Privacy-Aware Calibration Module, DP Training Wrapper, Privacy Budget Accountant (embedded), and Privacy Audit Interface (embedded). Novel contribution framework validated. Group handoff documents delivered. Ready to proceed to Phase 3 mass generation upon receipt of CARLA synthetic dataset batches from IT22541284.

Supervisor's comments.

Student progressing well in research project: Yes / No

If NO:

Project work to be carried out before next evaluation period:

Signature: 

Date: 30/042026

Date: 1st May - 13th May

Individual Component: Privacy Integration: Implementing the epsilon-differential privacy enforcement pipeline including the Privacy-Aware Calibration Module, DP Training Wrapper, Privacy Budget Accountant, HMAC-SHA256 Certificate Signing security component, and novel epsilon calibration attack study. Integration with IT22541284 real CARLA calibration data completed in this period.

Summary of RP work carried out during the reporting period

Date	RP Work
01/05/2026	Received calibration_stats_yubitha_final.json from IT22541284 containing real CARLA runtime measurements across five towns. File includes per-town vehicle densities, pedestrian densities, road lengths, intersection counts, weather distributions, and time-of-day distributions. Analyzed the file structure and confirmed all seven required fields are present in the correct format. Identified that the LaplaceCalibrationModule only extracts the seven aggregate fields and discards all per-town breakdowns, satisfying GDPR Article 5(1)(c) data minimization principle.
05/05/2026	Set up the new clean project environment (research conda environment, Python 3.11.15). Ran the LaplaceCalibrationModule against calibration_stats_yubitha_final.json. Module applied Laplace noise to all seven aggregate fields at epsilon_calibration = 1.8 and produced calibration_stats.json and calibration_audit_log.json. Ran the full DP training pipeline across all three ablation configurations: Config A final epsilon 0.9921 with sigma 2.5977, Config B final epsilon 2.9988 with sigma 1.1938, Config C final epsilon 7.9942 with sigma 0.7498. All three runs completed 250 steps with budget_respected = True and halted_early = False. Ran the MIA calibration study and all four configurations returned attack success rates near 50% confirming the privacy guarantees are held on the current dataset.

08/05/2026	Online progress review meeting with co-supervisor Mr. Dinith Primal. Demonstrated the full Privacy Integration pipeline live via screen share: ran run_calibration.py against the real CARLA data from IT22541284 showing Laplace noise applied to each field, ran run_training.py showing DP-SGD training with RDP budget tracking across epochs, and ran the full unit test suite confirming 50 tests passing with 0 failures. Co-supervisor reviewed the audit report and budget log outputs. Progress confirmed satisfactory for Phase 2 completion.
11/05/2026	PP1 Progress Presentation delivered in person. Presented the Privacy Integration component to the panel covering the four sub-modules, user requirements FR-01 through FR-07, the novel epsilon calibration study as the research contribution, and the HMAC-SHA256 certificate signing security component. Panel noted that IT22110220 and IT22541284 did not have dedicated security components in their individual parts. Both members subsequently added security components as a resolution: IT22541284 integrated AES-256 encryption, SHA-256 hashing, and RBAC access control into the dataset generation pipeline; IT22110220 implemented a Data Poisoning Detection mechanism that runs statistical checks on label distributions before model training begins.
14/05/2026	Added HMAC-SHA256 certificate signing as a dedicated security component in audit/certificate_signing.py. The CertificateSigner class signs every audit report after generation and produces a tamper-evident certificate. Any modification to the audit report after signing causes verification to fail, satisfying GDPR Article 5(2) accountability requirements. Integrated signing into run_training.py so every ablation config automatically produces a signed certificate alongside the audit report. Verified all 50-unit tests continue to pass including 7 new tests covering certificate generation and tamper detection. Finalized GitHub repository structure and .gitignore for the Privacy-Integration component.

Supervisor's comments.

Student progressing well in research project: Yes / No

If NO:

Project work to be carried out before next evaluation period:

Signature: 

Date: 16/05/2026

Date: 17th May - 27th May

Individual Component: Privacy Integration: Phase 3 commenced. Preparing the DP training pipeline for real CARLA synthetic dataset. Dataset combining approach confirmed with IT22541284. Phase 3 training scripts updated and ready pending dataset delivery.

Summary of RP work carried out during the reporting period

Date	RP Work
17/05/2026	Phase 3 commenced for the Privacy Integration component. Conducted group coordination meeting to confirm Phase 3 plan. Agreed that IT22541284 will generate synthetic data across five CARLA towns (Town01 to Town05) using the privatized calibration_stats.json delivered at the end of Phase 2. Confirmed that all five town outputs will be combined into a single merged dataset for DP training runs. Confirmed calibration_stats.json with calibration_epsilon_spent = 1.8 is the correct file for IT22541284 to use for all five towns and all sensor types including RGB, depth, and semantic segmentation.
20/05/2026	Clarified dataset usage with IT22541284. Confirmed that the same calibration_stats.json values apply uniformly across all five towns and all sensor configurations. Began preparing the Phase 3 data loading pipeline. Created prepare_dataset.py to merge the five individual town dataset files into a single combined dataset for DP training. The script uses torch.cat to concatenate camera_frames, lidar_frames, and labels tensors across all five towns and saves the merged output as combined_camera_frames.pt, combined_lidar_frames.pt, and combined_labels.pt. Awaiting delivery of town dataset files from IT22541284 to complete the merge step.

23/05/2026	Updated run_training.py to support real CARLA synthetic dataset loading for Phase 3. Replaced the toy synthetic data loader (random tensor generation) with a real dataset loader that reads the combined .pt tensor files produced by prepare_dataset.py. The updated loader accepts the combined_camera_frames.pt and combined_labels.pt files and wraps them in a DataLoader compatible with the existing DPTrainingWrapper. All three ablation configurations (Config A epsilon 1.0, Config B epsilon 3.0, Config C epsilon 8.0) will train on the same combined dataset. Phase 3 training pipeline is ready and awaiting dataset delivery from IT22541284.
25/05/2026	Reviewed Phase 3 responsibilities and confirmed deliverables for IT22066916. After DP training completes, the following outputs will be delivered: three audit_report.json files (one per ablation config), three audit_certificate.json files signed with HMAC-SHA256, three budget_log.csv files with per-step epsilon values, and calibration_curve_data.json from the MIA calibration study. Verified that the full pipeline chain (run_calibration, run_training, run_mia_study) runs end to end without errors on development hardware in preparation for the real dataset runs.
27/05/2026	Continued monitoring Phase 3 dataset generation progress with IT22541284. Dataset files across all five towns are still being generated. Phase 3 training pipeline code is fully prepared and tested on development data. All unit tests remain at 50 passed with 0 failures. Upon receipt of the five town datasets from IT22541284, the immediate steps are to run prepare_dataset.py to merge all towns, run run_training.py to execute DP training across all three ablation configurations, run run_mia_study.py to produce the epsilon calibration curve, and deliver signed audit reports to IT22066916 for compliance review.

Supervisor's comments.

Student progressing well in research project: Yes / No

If NO:

Project work to be carried out before next evaluation period:

Signature: 

Date: 01/06/2026

Date: 22nd July - 31st July

Individual Component: Privacy Integration: Phase 3 dataset acquisition and verification. Received and decrypted the first synthetic dataset delivery from IT22541284, verified calibration provenance and dataset integrity, identified missing ground-truth annotations, rebuilt the development environment following a machine reset, and completed a full codebase review identifying structural and security defects for remediation.

Summary of RP work carried out during the reporting period

Date	RP Work
22/07/2026	Received the first synthetic dataset delivery from IT22541284, an 8GB archive containing five CARLA towns (Town01 through Town05) with 2000 frame sets per town. Each frame set comprises an RGB camera image, a depth map and a semantic segmentation image, all encrypted with AES-256. Reviewed the accompanying metadata files: town_manifest.json holding a SHA-256 digest per file, town_metadata.csv recording weather category, time of day and event type per frame, and town_config_metadata.json describing the scene configuration used. Confirmed that the calibration_profile block in Town01's config metadata matches calibration_stats.json exactly, including vehicle_density_per_km of 2.5139 and calibration_epsilon_spent of 1.8, verifying that IT22541284 configured CARLA using the privatized statistics rather than raw Waymo values.
23/07/2026	Requested the AES-256 decryption key and the supporting security modules from IT22541284, since the dataset could not be read without them. Received security_engine.py and rbac_gateway.py and reviewed the encryption scheme, confirming AES-256 in CBC mode with a 16-byte initialization vector prepended to each ciphertext and PKCS7 padding. Wrote decrypt_dataset.py as an independent implementation rather than reusing IT22541284's script, because her version deletes the encrypted source file after decryption, which would have made a partial or failed run unrecoverable across 8GB of data. The new implementation leaves sources untouched and verifies every decrypted file against its manifest digest before writing. Also identified a path bug in her loader script, which searched for town02_manifest.json inside the town01 folder, and reported it to her.

24/07/2026	Ran the decryption across all five towns. All 30,000 files were decrypted successfully with zero failures, and every file matched its SHA-256 manifest digest, confirming both that the key and scheme were correct and that no corruption occurred during archive transfer. Inspected the decrypted outputs and identified a significant gap: the metadata CSV contains only scene-level information such as weather and time of day, with no object-level ground truth. No bounding box annotations for vehicles, pedestrians or cyclists were present anywhere in the delivery.
25/07/2026	Raised the missing ground truth with IT22541284, noting that the absence of bounding boxes blocks both the privacy component and IT22110220's utility evaluation, since neither a classifier nor an object detector can be trained without object-level labels. Also requested confirmation that all five towns used the same calibration file, since only Town01 had been verified, along with the semantic segmentation class mapping, since the configuration specifies the CityScapes palette without stating how those classes map to the project's own car, pedestrian and cyclist categories. Queried the frame numbering, which begins at f003568 rather than f000000, to establish whether earlier frames were discarded as simulator warm-up or withheld.
26/07/2026	Rebuild the complete development environment following a full reset of the development machine. Installed Anaconda, created the research environment with Python 3.11.15, and installed PyTorch 2.5.1 with the CUDA 12.1 build to match the RTX 3050 Ti and its CUDA 12.5 driver. Installed Opacus 1.4.1, pycryptodome for the dataset decryption, and the remaining project dependencies from requirements.txt. Verified GPU availability through torch.cuda.is_available and confirmed the full unit test suite passed on the rebuilt environment, establishing that no project code was lost in the reset.

29/07/2026	Conducted a full review of the codebase in preparation for the next development phase. Identified several structural weaknesses that would become problems as the project grew: seventeen sys.path manipulation statements substituting for a proper package structure, duplicated logic between the training and attack runner scripts with one runner importing directly from another, configuration accessed as untyped dictionaries with no validation, and output filenames keyed only on the configuration name so that repeating an experiment under a different random seed would silently overwrite earlier results. Also identified two security defects: the AES dataset key and the HMAC audit signing key were both hardcoded in source files in a repository intended to be pushed to GitHub, which would place credentials permanently in version control history.
31/07/2026	Planned the remediation of the issues identified in the code review, prioritizing the two hardcoded credentials as the most urgent given the pending repository push, followed by the seed-aware output paths to prevent further loss of experimental results. Reviewed the existing test coverage and confirmed that while the four original privacy modules were well covered, the newer dataset handling, training and decryption code had no automated tests at all. Scoped the work required to bring the codebase to a standard suitable for external review, covering package structure, credential handling, configuration validation and test coverage.

Supervisor's comments.

Student progressing well in research project: Yes / No

If NO:

Project work to be carried out before next evaluation period:

Signature: 

Date: 01/08/2026

Date: 01st August - 31st August

Individual Component: Privacy Integration: Phase 3 experimental study. Diagnosed and corrected a label quality fault that prevented model training, rebuilt the data pipeline and codebase, completed the full experimental study across three independent seeds with statistical testing, generated the signed privacy certificate, and delivered the complete artefact set for independent compliance verification.

Summary of RP work carried out during the reporting period

Date	RP Work
02/08/2026	Received the updated synthetic dataset from IT22541284, now including the object-level ground truth that was missing from the first delivery. The new package contains YOLO-format bounding box label files and single-channel class segmentation masks in addition to the RGB, depth and semantic frames. Updated the decryption script to recurse into the new class_masks subfolder and to copy the unencrypted label files verbatim, noting that an empty label file is a valid state meaning no objects were detected in that frame rather than an error. Decrypted all 40,000 image files across the five towns with zero failures, every file matching its SHA-256 manifest digest, and copied 10,000 label files.
03/08/2026	Began training on the real dataset for the first time. Wrote real_dataset.py to load the decrypted frames and derive a binary pedestrian or cyclist presence label from the bounding box files, and a DP-compatible convolutional model using GroupNorm rather than BatchNorm since Opacus does not support the latter. Initial runs completed without error, but the model failed to learn: accuracy remained frozen at exactly the negative class proportion across every epoch, indicating the model had collapsed to always predicting the majority class rather than learning the task.

04/08/2026	Conducted a systematic investigation into the training failure, covering five learning rates, two optimizers, two batch sizes and three class-imbalance strategies. Three genuine findings emerged and are now documented in the code. First, the Adam optimizer destabilizes under DP-SGD because its second-moment estimate accumulates the injected privacy noise, so training was switched to SGD with momentum, which is the standard choice in the DP-SGD literature. Second, loss-based class reweighting destabilizes under per-sample gradient clipping, since clipping removes the intended magnitude correction while retaining directional bias. Third, a weighted random sampler is silently discarded by Opacus during make_private, because its privacy accounting requires Poisson subsampling, so class rebalancing had to be implemented by physically duplicating minority-class indices instead.
05/08/2026	Identified the actual root cause of the training failure, which none of the hyperparameter changes had addressed. Built a label quality diagnostic to measure the bounding box size distribution of pedestrians and cyclists at the training resolution then in use. The diagnostic found that 62.3 percent of frames labelled positive had their largest pedestrian or cyclist smaller than approximately 8 by 8 pixels after resizing. Those labels were asserting the presence of objects the network could not physically resolve, meaning most of the positive class was effectively label noise. It also found that the 800 by 600 native capture was being resized to a 256 by 256 square, distorting every object by 3.12 times horizontally against 2.34 times vertically.

06/08/2026	Rebuild the data pipeline to correct the faults identified by the diagnostic. Resolution changed to preserve the native four to three aspect ratios, so objects are no longer distorted and are substantially larger at training time. Introduced a three-way label rule in which frames whose only pedestrian or cyclist falls below a minimum visible size are excluded from the dataset rather than labelled negative, following the convention used by the KITTI benchmark for objects below its minimum height threshold. Reduced network down sampling and replaced average pooling with max pooling so that small objects survive to the classifier rather than being averaged away. Added ImageNet normalization. Validated the fix: the non-private baseline reached a test AUC of 0.84, up from approximately 0.50, confirming the task is genuinely learnable and that the earlier failure was a preprocessing fault rather than a limitation of differential privacy.
08/08/2026	Addressed methodological gaps in the experimental setup before running the full study. Introduced stratified train, validation and test splits, since all previous loss figures had been measured on training data with no held-out set. Added per-epoch validation metrics covering accuracy, precision, recall, F1 and AUC, with AUC adopted as the primary metric because the rebalanced training distribution differs from the natural test distribution and makes accuracy misleading at a fixed threshold. Added model checkpointing and full random seeding across Python, NumPy and Torch with cuDNN autotuning disabled, satisfying the reproducibility requirement stated in the Design Document.
11/08/2026	Completed a professional rebuild of the codebase. Removed two hardcoded credentials, the AES dataset key and the HMAC audit signing key, replacing them with environment-based loading and adding a regression test that fails if either literal ever returns to the source. Made output paths name spaced by random seed, correcting a defect under which repeating a configuration under a different seed silently overwrote the earlier result. Replaced untyped dictionary configuration access with validated settings that reject contradictory values at load time, eliminated seventeen module path manipulations by making the project a proper installable package, and expanded the test suite from 50 to 117 tests covering the label rules, data splits, metrics, decryption and credential handling.

13/08/2026	Executed the full experimental study across three independent random seeds, each covering the non-private baseline and the three privacy configurations at epsilon 1.0, 3.0 and 8.0, giving twelve training runs and their corresponding membership inference attack studies. Verified reproducibility empirically by repeating runs at a fixed seed and obtaining identical results down to the per-epoch loss and the final test AUC. Results: the non-private model averaged 0.83 test AUC with an attack success rate of 56.7 percent (attack AUC 0.64), while all three private configurations averaged 0.62 to 0.67 test AUC with attack success rates of 50.3 to 51.6 percent, indistinguishable from random guessing.
18/08/2026	Performed paired statistical testing across the three seeds. The difference between protected and unprotected configurations is statistically significant on both the utility cost and the attack resistance. The differences between the three privacy strengths are consistent in direction across all three seeds but do not reach significance at three replications and are reported as such rather than overstated. Generated the signed privacy certificate aggregating all runs, which declares four known limitations including the disabled secure random number generator and an unresolved question regarding how the calibration privacy budget composes across seven separate queries.
22/08/2026	Packaged and delivered the complete artefact set to IT22066916 for independent compliance verification: 51 files comprising audit reports, signed certificates, per-step privacy budget logs, model checkpoints, attack results and both the privatized and unprivatized calibration statistics, accompanied by a SHA-256 manifest so the handoff itself can be integrity-checked. The accompanying documentation discloses the calibration composition question and the other known limitations rather than leaving them to be discovered. Prepared the repository structure for the shared GitHub repository, verifying before any commit that no credentials, unprivatized source statistics, model weights or generated outputs would be included.

Supervisor's comments.

Student progressing well in research project: Yes / No

If NO:

Project work to be carried out before next evaluation period:

Signature: 

Date: 31/08/2026

Date: 01st September - 30th September

Individual Component: Privacy Integration: Phase 4 integration and verification. Supported the independent compliance verification of the delivered artefacts, corrected the reported calibration privacy budget, moved the component into the shared GitHub repository with continuous integration, and joined it to the end-to-end pipeline with the other three components.

Summary of RP work carried out during the reporting period

Date	RP Work
02/09/2026	Met with IT22066916 to walk through the handoff package delivered on 22/08/2026. Explained the layout of the signed reports, the per-step privacy budget logs and the SHA-256 manifest, and how each file can be checked without trusting the Privacy Integration code. Confirmed that the manifest matched every received file before the verification work began.
07/09/2026	Responded to a verification finding from IT22066916 on the calibration statistics. The independent composition check showed that the seven calibration queries, each run at epsilon 1.8, compose to a total budget of 12.6 under basic composition, while the earlier report stated the total as 1.8. Confirmed the finding, corrected the reporting code and regenerated the signed calibration report. The added noise and the released statistics were not changed, so no downstream dataset had to be regenerated. The limitation is now disclosed in the privacy certificate.
10/09/2026	Resolved a model loading problem raised by the Risk and Compliance attack scripts. The saved checkpoints held the full training state, including the epsilon value and settings, and the first attack script assumed a fully connected network. Shared the exact convolutional model class with group normalization and a helper that extracts only the weights, so the attack could load every private and non-private model with the correct architecture.

15/09/2026	Moved the Privacy Integration package into the shared group repository. Defined the JSON Schema contracts for every file the component exchanges with the other components, added the shared library for key loading and report signing, and confirmed that no credentials, unprivatized statistics, model weights, encrypted data or generated outputs were included in any commit.
19/09/2026	Set up continuous integration for the shared repository using GitHub Actions. Every push now runs the linting checks and the full test suite of each component. Added pre-commit hooks that block committed keys and environment files, and kept the regression test that fails if an old AES or HMAC key returns to the source.
24/09/2026	Built the pipeline stages for Privacy Integration so that calibration, decryption, private training, the membership inference attacks and certificate generation each run as a separate command-line stage. Joined these stages with the stages of the other three components and contributed to the end-to-end pipeline test, which runs the full flow on a small sample on every push.
29/09/2026	Addressed the comments from the progress review panel. Added a positive control to the privacy evaluation by confirming that the same membership inference attack succeeds against the non-private baseline, so that a near-chance result on the private models can be trusted. Updated the component documentation and the privacy certificate to record the review comments and the changes made in response.

Supervisor's comments.

Student progressing well in research project: Yes / No

If NO:

Project work to be carried out before next evaluation period:

Signature: 

Date: 30/09/2026

Date: 01st October - 18th October

Individual Component: Privacy Integration: Phase 5 documentation and dissemination. Completed the individual final report, contributed to the common integrated solution report and the project website, and prepared the research paper and the final presentation.

Summary of RP work carried out during the reporting period

Date	RP Work
02/10/2026	Completed the individual final report for the Privacy Integration component. The report covers the calibration mechanism, the DP-SGD training study across three seeds, the membership inference results, the statistical testing, the signed privacy certificate and the known limitations, including the corrected calibration budget.
06/10/2026	Contributed the Privacy Integration sections to the common integrated solution report, covering the shared technology stack, the interface contracts, the problems found during integration and how each was fixed. Checked every number in the common report against the signed reports and budget logs so that the individual and group documents agree.
09/10/2026	Worked on the project website with the group. Replaced the earlier website folder in the shared repository with the updated version, which publishes the proposal, progress and final documents of all four members. Checked that the public site contains no credentials, unprivatized statistics or other private data before it was deployed through GitHub Pages.
13/10/2026	Started preparing the group research paper for submission to a conference. Shortlisted suitable venues with submission deadlines before December, compared their scope and indexing, and drafted the outline covering the integrated pipeline, the privacy and utility results and the independent verification.

16/10/2026	Prepared for the final presentation. Drafted the Privacy Integration slides on the calibration mechanism, the DP-SGD results and the membership inference evaluation, and rehearsed the demonstration of the pipeline stages, the budget log and the certificate verification.
17/10/2026	Fixed the calibration privacy budget. Calibration now enforces a total of epsilon 1.8, split evenly over the seven queries (0.257 each), and the independent composition check by IT22066916 passes on the new outputs; the dataset recapture at the corrected budget is pending. Answered the PP2 panel comment with a stress test: models trained on 500 frames for 60 epochs so the non-private control leaks (attack AUC 0.628 over five seeds), while DP-SGD reduced it to 0.501 at epsilon 1, significant at every budget (p at most 0.001), with stronger privacy giving a weaker attack.
18/10/2026	Ran a stronger, reference-calibrated membership inference attack on the stress-test models over three seeds. Without DP it caught 7.9 percent of members at a 1 percent false-positive rate; at epsilon 1 this fell to 1.1 percent, which is chance. Added signed one-page PDF certificates for Privacy Integration and Risk and Compliance, each carrying a SHA-256 fingerprint and an HMAC signature and prepared the final presentation.

Supervisor's comments.

Student progressing well in research project: Yes / No

If NO:

Project work to be carried out before next evaluation period:

Signature: 

Date: 18/10/2026